

Enterprise Cloud Modernization & Azure to AWS Migration Case Study – Zerodha

Executive Summary

Our organization, a leading provider of cloud, data engineering, analytics, and modernization solutions, partnered with Zerodha to execute a large-scale Azure-to-AWS migration program. The engagement focused on migrating virtual machines, Spark workloads, Apache Airflow orchestration platforms, Kafka streaming infrastructure, operational databases, and analytics environments. The objective was to improve scalability, governance, performance, reliability, and cost efficiency while building a future-ready data platform capable of supporting rapidly growing trading, customer analytics, risk management, and market intelligence workloads.

Business Challenges

- Rising Azure infrastructure costs across VM estates, Spark clusters, storage, and streaming platforms.
- Growing trading volumes generating terabytes of market and customer data daily.
- Complex dependencies across Airflow DAGs, Kafka topics, Spark jobs, APIs, and databases.
- Need for stronger governance, monitoring, disaster recovery, and security controls.
- Requirement to support future AI, machine learning, and advanced analytics initiatives.
- Limited operational visibility across distributed workloads.

Source Environment

Compute Layer: Azure Virtual Machines hosting applications, APIs, middleware, and reporting services.

Processing Layer: Azure Databricks and Spark workloads executing market analytics, ETL, and reporting jobs.

Streaming Layer: Kafka workloads handling market events, customer activities, and downstream integrations.

Orchestration Layer: Apache Airflow managing hundreds of DAGs and data pipelines.

Storage Layer: ADLS Gen2 and Azure Blob Storage.

Database Layer: PostgreSQL and MySQL environments supporting operational workloads.

Migration Approach

1. Discovery and dependency assessment.
2. AWS landing zone setup with security and governance controls.
3. Historical data migration from ADLS to Amazon S3 using AWS DataSync.
4. Database migration using AWS DMS to Amazon Aurora PostgreSQL and Amazon RDS.
5. Rehosting Azure VMs to Amazon EC2 using AWS Application Migration Service.
6. Migration of Spark workloads to Amazon EMR with performance optimization.
7. Migration of Kafka workloads to Amazon MSK.
8. Migration of Apache Airflow to Amazon MWAA.
9. Validation, reconciliation, performance testing, and phased production cutover.
10. Post-migration optimization and operational stabilization.

Target AWS Architecture

Data Lake: Amazon S3, AWS Glue Catalog, AWS Lake Formation.
Processing: Amazon EMR, Apache Spark, AWS Glue.
Streaming: Amazon MSK, Kafka Connect, Kafka Streams.
Orchestration: Amazon MWAA.
Compute: Amazon EC2, Amazon ECS, Amazon EKS.
Databases: Amazon Aurora PostgreSQL, Amazon RDS PostgreSQL, Amazon RDS MySQL.
Security: IAM, KMS, Secrets Manager, Lake Formation.
Monitoring: CloudWatch, CloudTrail, AWS Config.
Analytics: Athena, QuickSight, Tableau, Power BI.
AI/ML: SageMaker and Bedrock.

Migration Validation Framework

Our team implemented an automated validation framework covering row-count reconciliation, schema validation, Spark output comparison, Kafka message validation, Airflow DAG execution checks, database consistency validation, performance benchmarking, and security compliance verification. This ensured business continuity with minimal disruption to trading operations.

Business Value Achieved

- 25–30% reduction in infrastructure costs.
- 40% improvement in Spark job execution performance on Amazon EMR.
- Improved scalability during peak market activity.
- Reduced operational overhead through managed AWS services.
- Enhanced governance, monitoring, and disaster recovery capabilities.
- Faster onboarding of analytics and AI use cases.

Impact Generated

Operational Efficiency: Reduced manual administration through managed services.

Performance & Scalability: Improved throughput for market analytics and reporting.

Reliability: Achieved greater than 99.9% migration validation accuracy.

Business Agility: Accelerated delivery of new data products and analytics initiatives.

Technology Stack

Amazon S3, Amazon EMR, Apache Spark, Amazon MSK, Apache Kafka, Amazon MWAA, Apache Airflow, AWS Glue, AWS DMS, AWS DataSync, Amazon EC2, Amazon ECS, Amazon EKS, Amazon Aurora PostgreSQL, Amazon RDS, AWS IAM, AWS KMS, AWS Lake Formation, AWS Secrets Manager, CloudWatch, Athena, QuickSight, Tableau, Power BI, SageMaker, and Amazon Bedrock.

Conclusion

The engagement successfully modernized Zerodha's cloud ecosystem and established a secure, scalable, and cloud-native AWS platform capable of supporting mission-critical trading, analytics, streaming, and AI workloads. By migrating Spark, Kafka, Airflow, databases, and virtual machine workloads, our company enabled lower costs, improved performance, stronger governance, and a future-ready foundation for innovation.